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To cite this article before publication: Amare Getnet G. Worku *et al* 2026 *Environ. Res. Commun.* in press <https://doi.org/10.1088/2515-7620/aeaad0>

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Variable Source Area (VSA) Modeling for Water Management: Streamflow Analysis in Upper Gilgel Abay Watershed, Ethiopia

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Abstract

Reliable streamflow simulation is fundamental for effective water resources management in data-scarce watersheds, particularly where runoff is dominated by saturation-excess processes that are inadequately represented by many conventional hydrological models. This study evaluates the applicability of the Variable Source Area (VSA) hydrological model for simulating streamflow dynamics in the Upper Gilgel Abay watershed, a humid tropical watershed in the Upper Blue Nile Basin of Ethiopia, an ecologically and socio-economically vital region. To improve parameter estimation, a two-stage calibration strategy was implemented, combining a Monte Carlo based Latin Hypercube Sampling (LHS)) for efficient exploration of the feasible parameter space and identification of behavioural parameter sets and Differential Evolution optimization (DEoptim) for refinement of the optimal parameter values. Model performance was evaluated using the Nash-Sutcliffe Efficiency (NSE), Kling-Gupta Efficiency (KGE), and Percent Bias (PBIAS). The calibrated VSA model achieved very good performance during calibration (NSE = 0.81, KGE = 0.86, PBIAS = -4.87%) and good performance during validation (NSE = 0.69, KGE = 0.75, PBIAS = 5.13%). The model successfully captured the seasonal variability of streamflow, though peak flows were generally overestimated and low flows were underestimated during the validation period. Parameter analysis revealed substantial differences in parameter influence and uncertainty, with the groundwater recession coefficient (rec_coef) identified as the dominant parameter, while available water capacity (AWCper) and Depth showed secondary influence. Storage-related parameters, particularly Depth and SATper, remained comparatively uncertain, indicating limitations in constraining internal watershed storage from streamflow observations alone. Overall, the results demonstrate that the VSA framework, combined with LHS-DEoptim calibration and complementary sensitivity and uncertainty analyses, provides a physically meaningful and computationally efficient approach

for representing saturation-excess runoff in data-limited Ethiopian highland watersheds, supporting hydrological evaluation and water management.

Keywords: Variable Source Area (VSA) model, streamflow simulation, saturation-excess runoff, LHS-DEoptim, parameter sensitivity, data-scarce watershed, Upper Blue Nile Basin, Ethiopia.

1. Introduction

The Upper Gilgel Abay watershed is located in the Upper Blue Nile Basin of Ethiopia and plays an important role for regional hydrology and water resource management. Dynamic water availability is essential for sustaining livelihoods and ecosystems across the basin (Wedajo et al., 2024), while effective water-resource management is increasingly important under climate variability and human-induced changes (Adem et al., 2016). Recent assessments have identified water stress in the eastern and northeastern cropland zones and slight stress in transitional agricultural areas of the Lake Tana Basin, highlighting the increasing need for improved understanding and management of water availability (Worku et al., 2026). Hydrological modelling provides an important framework for understanding watershed processes and supporting water-resource planning and management (Mashaly & Fernald, 2020). Hydrological models are simplified representations of hydrological processes developed for both process understanding and prediction (Xu, 2002), ranging from empirical and conceptual approaches to physically based distributed models (Refsgaard et al., 2022).

Among conceptual rainfall-runoff approaches, the Variable Source Area (VSA) framework provides a process-oriented representation of runoff generation by accounting for the spatial and temporal variability of runoff-contributing areas. This framework is particularly relevant to watersheds where saturation-excess runoff is an important runoff-generation mechanism, because the contributing area expands and contracts in response to changing soil-moisture conditions. VSA-based modelling can also represent the effects of soil properties, land cover, and topographic characteristics on the spatial distribution of runoff-generating areas (Amin et al., 2017; Kumar et al., 2016; Panjabi et al., 2020). Consequently, the framework has potential applications beyond streamflow prediction, including watershed management and assessment of nonpoint-source pollution in saturation-excess dominated catchments (Chang et al., 2021).

Previous hydrological studies in the Upper Gilgel Abay watershed have predominantly employed models such as the Soil and Water Assessment Tool (SWAT) and the Hydrologiska Byråns Vattenbalansavdelning (HBV) model (Bizuneh et al., 2021). These studies have contributed substantially to understanding watershed hydrology and water resources management. However, their primary emphasis has been on simulating streamflow rather than explicitly evaluating the saturation-excess runoff processes that dominate humid Ethiopian highlands. Consequently, relatively little attention has been given to assessing whether a variable source area framework provides a more hydrologically appropriate representation of runoff generation in Sub-Saharan Africa (Banda et al., 2022).

The Upper Gilgel Abay watershed is characterized by shallow soils, frequent intense rainfall and pronounced seasonal changes that make the runoff generation process dominated by saturation excess runoff processes. These conditions promote runoff production through variable source areas. Despite these hydrological characteristics, applications of semi-distributed VSA models remain scarce within the Upper Blue Nile Basin. Consequently, the capability of the VSA framework to simulate streamflow dynamics and improve understanding of runoff generation mechanisms under data-scarce tropical conditions remains largely unexplored (Keshta et al., 2019).

Accordingly, this study addresses this knowledge gap by providing the first application and comprehensive evaluation of the Variable Source Area (VSA) hydrological model in the Upper Gilgel Abay watershed. Unlike previous studies that primarily assessed predictive performance using SWAT or HBV models, this study investigates whether a hydrological framework explicitly representing saturation-excess runoff can realistically simulate daily streamflow while simultaneously improving understanding of the dominant runoff generation processes. To enhance parameter estimation and reduce parameter uncertainty, a two-stage calibration strategy was adopted, combining a Monte Carlo based Latin Hypercube Sampling (LHS) for efficient exploration of the feasible parameter space and identification of behavioural parameter sets with Differential Evolution optimization (DEoptim) for refinement of the optimal parameter values. Parameter uncertainty and identifiability were subsequently evaluated from the behavioural simulations, while a variance-based Sobol analysis was used to quantify parameter sensitivity and interaction effects.

Therefore, the objective of this study is to evaluate the applicability of the Variable Source Area hydrological model for simulating daily streamflow in the Upper Gilgel Abay watershed, Ethiopia. Specifically, the study aims to: (1) Implement a two-stage Latin Hypercube Sampling (LHS) and Differential Evolution optimization (DEoptim) calibration framework to efficiently explore the feasible parameter space, identify behavioural parameter sets, and determine an optimum parameter combination; (2) Evaluate VSA model performance in simulating daily streamflow during calibration and independent validation using complementary statistical performance measures; and (3) Assess parameter uncertainty, identifiability, and global sensitivity, and evaluate the suitability of the VSA framework for representing saturation-excess runoff generation under data-scarce Ethiopian highland conditions.

2. Materials and Method

2.1. Study area description

The Upper Gilgel Abay watershed has an area of 1632 km² at the gauging station near Wetet Abay town and with an elevation range of 1889 to 3524 meters above sea level is located in the Lake Tana sub-basin of the Upper Blue Nile Basin (Fig 1). It is the largest of the four main watersheds (Gilgel Abay, Gumara, Megech and Rib) that join Lake Tana, and the principal source of flow to Lake Tana (Rientjes et al., 2011). It has geographical coordinates with the latitudes and longitudes between 10° 56' 53 " - 11° 21' 58"N and 36° 49' 29" – 37° 23' 34" E (Adem et al., 2016) . The southern part of the watershed is mountainous, west and southeast periphery of the watershed has undulated topography, while the remaining part is a low laying plateau with a gentle slope. The geology is composed of outcrops of Tertiary, Quaternary volcanic rocks and alluvial deposits.

Haplic Luvisol (57.2%) and Haplic Alisol (39.4%) are the two most common soil types of the watershed and predominantly covered by cultivated land, grassland, and shrubland with shrinking forest cover land (Abuhay et al., 2023) . Monsoon agriculture is the most significant economic activity in the watershed. The rainy season (June-September) and the dry season (October-May) are the two seasons (Alemseged & Tom, 2015) , which are influenced by the seasonal migration of the Inter-Tropical Convergence Zone (ITCZ) (Conway, 2000). The climate data obtained from the Ethiopia National Meteorological Agency (NMA)

(<http://www.ethiomet.gov.et>), Addis Ababa, Ethiopia. The watershed's annual average rainfall is 1624 to 1986 mm and about 78% of its rainfall occurs between June and September. The mean annual maximum and the minimum temperatures of the watershed were 25°C and 9°C, respectively.

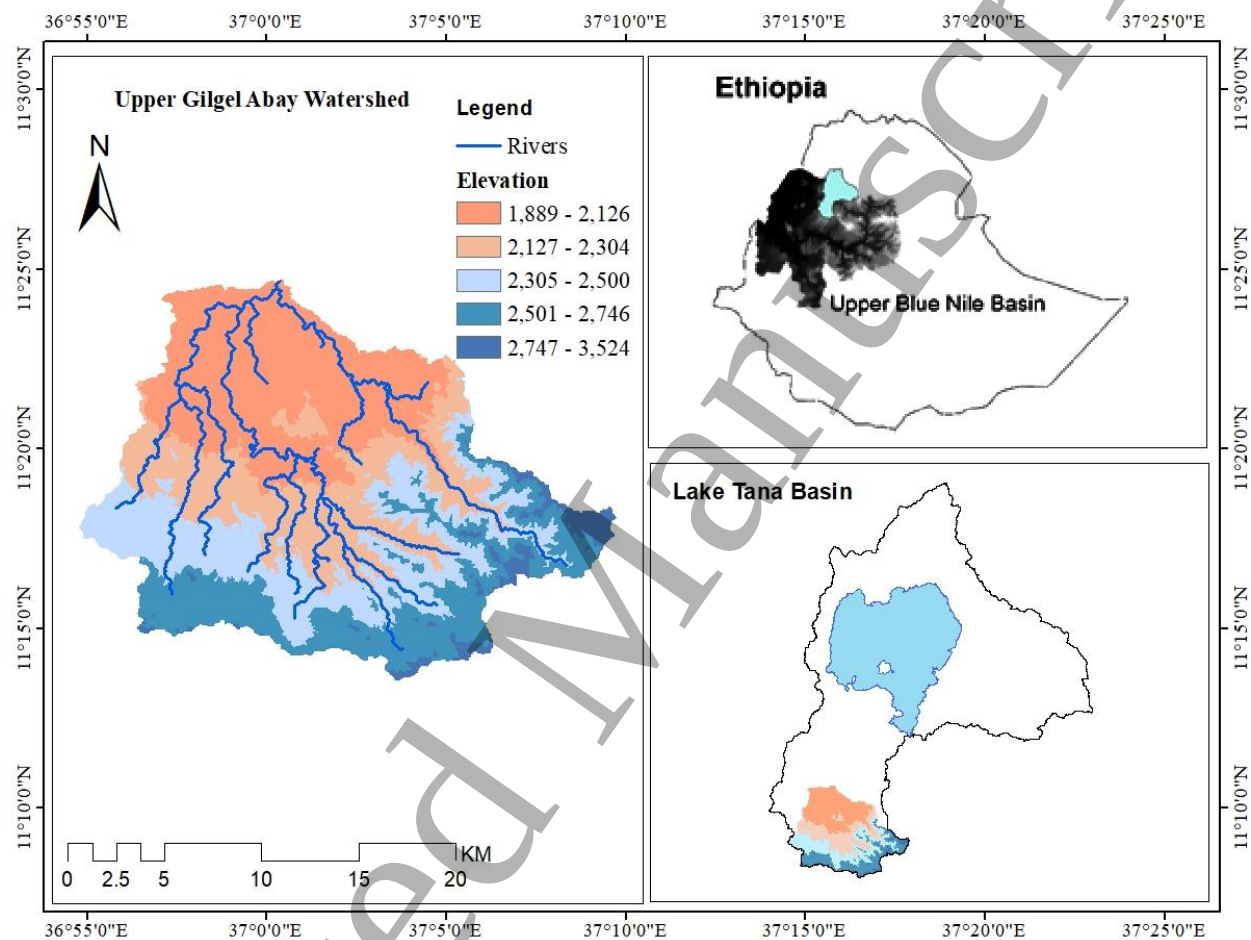


Figure 1: Location map of the Upper Gilgel Abay watershed in Ethiopia

2.2. Data sources and processing

Climate data - Daily rainfall (1994 - 2009) from four stations (Dangila, Injibara, Sekela, and Wetet Abay) were obtained from the Ethiopian National Meteorological Agency. There were few gaps in the daily rainfall and streamflow data. In the case of rainfall data, the number of gaps was less than 3% of the total observations, and for streamflow data, it was about 4.2% of the total daily observations. There was no missing data gap of more than five days. Missing data were imputed based on the mean-day approach where each missing data value was substituted

with the average value of all data available for that particular day (day of the month) of other years of the series. This technique was used considering the low percentage of missing data and the absence of continuous gaps, which made it unlikely for the imputation process to have any significant effect on the results (Teegavarapu, 2014).

Nevertheless, mean-day imputation could lower inter-annual variations and reduce the magnitude of extreme events. Other data imputation techniques, such as regression models, inverse distance weightings, and stochastic interpolation, could be considered. However, due to the short duration of missing data records and the lack of quality covariates, the use of these imputation procedures was deemed unnecessary. The areal rainfall was calculated using the Thiessen polygon approach (Thiessen, 1911) from the meteorological stations. Based on the rainfall distribution at the weather station, the Thiessen polygon is used to compute the average rainfall by linking neighbouring stations to build triangles by forming the vertical bisector of each side of the lines. Weighting the different stations as shown below in Table1 with their corresponding area yields the spatial average precipitation, which is given by

$$P = \frac{1}{A} \sum_{i=1}^n A_i P_i \dots \dots \dots \text{Eq1}$$

Where P is the areal average precipitation over the entire basin, n is the number of rain gauge stations and A is the total surface area of the catchment, which is the sum of sub-areas (A_i). The daily rainfall for the study region is interpolated using the data from the point stations.

The potential evapotranspiration (PET) was determined using the air temperature-dependent method developed by Enku and Melesse (2014), which was developed and evaluated specifically for data-scarce conditions in Ethiopia. The method has demonstrated good performance compared with the FAO-56 Penman–Monteith (PM) method across diverse hydro-climatic settings in the country (Adem, 2017), while further evaluations have confirmed its robustness when the temperature exponent is appropriately calibrated (Mengistu & Amente, 2020). Therefore, the Enku and Melesse (2014) method was considered an appropriate and empirically supported approach for estimating PET in the present study, given the limited availability of complete meteorological observations during 1994-2009.

Streamflow - Daily observed stream flow data were obtained from the Ministry of Water, Irrigation & Energy of Ethiopia. Discharge measurement was carried out by recording the stage, cross-section of the river channel, and flow velocity at the gauging stations of the study catchment. The discharge records from 1994 to 2009 were used to model fittings of calibration and validation periods.

Table 1: Thiessen weight values for the stations

Stations	% contribution
Wetet Abay	0.29
Dangila	0.12
Injibara	0.21
Sekela	0.38

3. Variable Source Area (VSA) hydrological model

3.1. Model description

The Variable Source Area (VSA) hydrological model relies on the saturation-excess runoff generation principle, which is the main mechanism of runoff generation in humid, tropical, and densely vegetated watersheds (Dunne & Black, 1970; Hewlett & Hibbert, 1967). The model was developed based on the Thornthwaite-Mather water budget and Soil Conservation Service Curve Number (SCS) approach for overland runoff (Dahlke et al., 2009). In contrast to Hortonian runoff, where excess runoff is generated by rainfall intensities exceeding the infiltration rates, saturation-excess runoff arises from the presence of a water table that rises to the land surface and turns excess rainfall into runoff, independent of its intensity. The key idea of the VSA approach is to recognize the dynamic nature of the runoff producing areas depending on antecedent moisture, topographic convergence, and aquifer processes.

The model adopts a conceptual approach to the water balance model where the dynamics of soil moisture are determined by the equation:

$$S_t = S(-1) + P_t - I_{a,t} - ET_t - Q_{s,t} - Q_{b,t} \dots \dots \dots \text{Eq2}$$

Where $S(t)$ is the soil water storage at time t (mm), P is the precipitation (mm), I_a is the initial abstraction (mm), ET is the evapotranspiration (mm), Q_s is the surface runoff (mm), and $Q_{b,t}$ is the baseflow (mm).

The model runs daily and uses daily data of precipitation and evapotranspiration as an input. Hydrological functioning of the watershed is represented through a number of storage components: interception, depression storage, soil water, saturation-excess runoff, and groundwater baseflow. Precipitation undergoes initial abstraction and the amount depends on daily precipitation and the initial abstraction coefficient (I_{a_coef}). Thus, the difference between daily precipitation and initial abstraction represents the part of precipitation that may be used for either infiltration or runoff formation.

The soil water storage process can be modeled as a one-layer reservoir whose total storage capacity depends on soil depth (Depth, mm) and soil available water capacity (AWC_{per}, fraction). Actual evapotranspiration can be modeled based on potential evapotranspiration and soil saturation level. Saturation excess runoff will occur when soil water storage surpasses the threshold level determined by the multiplication between soil depth and soil saturation percent (SAT_{per}, fraction). This source area variable will increase along with soil moisture content and attain its peak point whenever the entire watershed is fully saturated. In addition to saturation excess runoff, direct runoff of impervious areas (Impervious, fraction) should be considered. All the water that fails to contribute to either surface runoff or evaporation will move down into a shallow groundwater reservoir, which will subsequently join the river flow as baseflow. Baseflow can be calculated through a linear reservoir model, controlled by a baseflow coefficient (BF1, mm/day) and recession coefficient (rec_{coef}, fraction). The initial soil water storage value (SW1, mm) determines the initial moisture level for simulation.

VSA modeling is especially appropriate for the Upper Gilgel Abay catchment because of the high rates of infiltration, strong seasonality of rainfall, shallow aquifers, and mainly agricultural land use in this area. VSA model parameters and their hydrological explanations are provided here in Table 2 (Schneiderman et al., 2007).

Table 2: VSA model parameters, initial values and sampling ranges

Parameter	Description	Unit	Min	Max
Depth	Average soil depth in a watershed	mm	50	2000
SATper	Porosity (fraction)	%	0	1
AWCper	Available Water Capacity (AWC) (fraction)	%	0	0.5
Impervious	Percent of the watershed that is impervious	%	0	0.01
C1	Curve Number storage coefficient	-	0	5.0
BF1	Base flow	mm/day	calculated	0.119
rec_coef	Recession coefficient	-	0.05	0.1
Se_min	unsaturated zone soil water is at or exceeds field capacity	mm	78	600
Ia_coef	Initial abstraction coefficient	%	0.04	0.2
C1	Curve number storage coefficient	-		
no_wet_class	Number of wetland classes used for wetland area classification	-	10	
Time to peak (Tp)	Time between rainfall peak and discharge peak	hour	calculated	6.27
Lat	Watershed latitude	degree	11.66	
α	Surface albedo	-	0.23	

3.2. Model calibration and validation

The available daily hydro-meteorological dataset (1994–2009) was divided into a calibration period (1994–2003; approximately 60% of the record) and an independent validation period (2004–2009; approximately 40%). A one-year warm-up period was used prior to calibration to minimize the influence of arbitrary initial model conditions. Consequently, this split-sample approach provides an independent assessment of model transferability and reduces the likelihood of overfitting (Klemeš, 1986). The feasible parameter ranges (Table 2) were defined based on published literature, previous applications of the VSA model, and hydrologically realistic limits representative of the Ethiopian highlands. Uniform probability distributions were assigned to all parameters because no prior information was available regarding their probability distributions.

Model calibration was performed using two-stage calibration strategy to improve parameter estimation while maintaining efficient exploration of the multidimensional parameter space. In the first stage, Latin Hypercube Sampling (LHS) (Owen, 1992), a Monte Carlo sampling technique that efficiently explores the multidimensional parameter space while ensuring uniform coverage of the predefined parameter ranges. A total of 10,000 parameter combinations were generated and evaluated, allowing comprehensive exploration of the feasible parameter space while minimizing sampling bias. Model performance for each simulation was evaluated using the Nash–Sutcliffe Efficiency (NSE). Following the performance classification proposed by (Moriassi et al., 2007), parameter sets achieving $NSE \geq 0.70$ were considered behavioural and were used to identify feasible parameter ranges, assess parameter identifiability, and parameter uncertainty.

In the second stage, the behavioural parameter ranges obtained from the LHS analysis were used to define narrower search bounds for Differential Evolution optimization (DEoptim) (Mullen et al., 2011). Differential Evolution is a population-based global optimization algorithm that iteratively improves candidate solutions through mutation, crossover, and selection operations. Thirty independent optimization runs were performed, each consisting of 200 iterations, to minimize the influence of stochastic initialization and improve convergence reliability. The parameter set yielding the highest calibration NSE was selected as the optimal solution for subsequent validation.

Model performance during calibration and validation was evaluated using multiple statistical performance measures, including the Nash-Sutcliffe Efficiency (NSE), Kling-Gupta Efficiency (KGE), Percent Bias (PBIAS), and the coefficient of determination (R^2). In addition to these quantitative statistics, model performance was assessed through hydrograph comparison, scatter plots, flow-duration curves, and GLUE-based prediction uncertainty analysis to evaluate the model's ability to reproduce seasonal streamflow dynamics, peak flows, recession characteristics, and low-flow behaviour.

Parameter identifiability was assessed using the behavioural parameter sets obtained during the LHS stage. Posterior parameter distributions, dot plots, and pairwise parameter relationships were analyzed to identify well-constrained parameters and evaluate parameter uncertainty. Parameters exhibiting relatively narrow behavioural ranges were considered well constrained by

the available streamflow observations, whereas broader behavioural distributions indicated greater uncertainty. Prediction uncertainty was further quantified using the Generalized Likelihood Uncertainty Estimation (GLUE) framework based on the behavioural parameter sets, with uncertainty expressed as the 90% prediction uncertainty (90PPU) interval.

3.2.1. Global sensitivity and parameter uncertainty analysis

A global sensitivity analysis was conducted using the variance-based Sobol method (Puy et al., 2022; Sobol, 2001) to quantify the relative influence of VSA model parameters on streamflow simulation performance. The analysis evaluated first-order sensitivity indices (S_1), representing the individual contribution of each parameter, and total-order indices (S_T), representing the overall contribution of each parameter including interaction effects. The analysis was performed over the specified prior parameter ranges using the model performance metric NSE during the calibration period. Confidence intervals for the sensitivity indices were estimated using bootstrap resampling. The Sobol analysis complements the behavioural-parameter and identifiability analyses by providing a global assessment of parameter influence and parameter interactions across the prescribed parameter space.

3.3. Model performance evaluation

For this study, three different model performance statistics: Nash-Sutcliffe efficiency (Nash & Sutcliffe, 1970) , percent bias (PBIAS), and Kling Gupta efficiency (KGE) (Gupta et al., 2009) were used to compare the simulated stream-flow against the observed ones.

The NSE is defined by the formula

$$NSE = 1 - \frac{\sum_{i=1}^n (Q_{O i} - Q_{S i})^2}{\sum_{i=1}^n (Q_{O i} - \overline{Q_o})^2} \dots \dots \dots \text{Eq3}$$

Where $Q_{O i}$ is the observed flow at the i^{th} day, $Q_{S i}$ is the simulated flow on the i^{th} day, $\overline{Q_o}$ is the mean of the observed flow and n is the total number of observations. An NSE value of 1 indicates perfect agreement between observations and modeling simulations.

Percent bias (PB) measures the average tendency of simulated values, which can either be larger or smaller than their observed ones, with an optimal value of 0. It can be calculated by

$$PBIAS = \frac{\sum_{i=1}^n (Q_{oi} - Q_{si})}{\sum_{i=1}^n Q_{oi}} \times 100 \dots \text{Eq4}$$

The Kling-Gupta efficiency is given by the formula

$$KGE = 1 - \sqrt{(r - 1)^2 + \left(\frac{\sigma_s}{\sigma_o} - 1\right)^2 + \left(\frac{\mu_s}{\mu_o} - 1\right)^2} \dots \text{Eq5}$$

Where r indicates the Pearson product-moment correlation coefficient between the simulated and observed values, σ_s is the standard deviation in simulations, σ_o is the standard deviation in the observations, μ_s is the simulation mean, and μ_o is the observation mean. KGE value of 1 shows perfect agreement between observations and simulations.

4. Results

4.1. Parameter calibration and uncertainty analysis

The first stage of calibration employed the Monte Carlo calibration using Latin Hypercube Sampling (LHS) generated 10,000 parameter combinations spanning the predefined parameter space. Applying the behavioural criterion ($NSE \geq 0.70$) resulted in a subset of behavioural parameter sets that satisfactorily represented the observed streamflow dynamics. These behavioural simulations were subsequently used to constrain the feasible parameter space and investigate parameter identifiability (Table 3).

In the second stage, the behavioural parameter ranges identified from the LHS analysis were used as search bounds for Differential Evolution optimization (DEoptim). This hybrid calibration strategy combines the efficient global exploration capability of Monte Carlo sampling with the robust local optimization performance of Differential Evolution, thereby reducing parameter uncertainty while improving model performance. The optimized parameter set achieved the highest calibration performance ($NSE = 0.811$) and was subsequently used for independent model validation. Comparison between the initial parameter ranges and the behavioural ranges derived from the LHS analysis demonstrates clear differences in parameter identifiability (Table 3). The groundwater recession coefficient (rec_coef) and available water capacity parameter (AWC_{per}) became substantially constrained during calibration, indicating that these parameters exert strong control on streamflow simulation. Conversely, the storage-related parameters

(Depth, SATper and Se_min) retained comparatively broad behavioural ranges, indicating considerable parameter uncertainty. These parameters compensate for one another in reproducing observed hydrographs, making their unique estimation difficult using streamflow observations alone.

Table 3: Behavioural parameter ranges identified using Latin Hypercube Sampling (LHS) and optimum parameter values using DEoptim for the VSA hydrological model

Model	Parameter	Units	Initial		Behavioural Range (LHS)		Optimized Value (DEoptim)
			Lower	Upper	Lower	Upper	
VSA	Depth	mm	30	2000	566.2788	1901.7174	690.518
	SATper	%	0	1	0.0395	0.2489	0.0657
	AWCper	%	0	0.5	0.1387	0.4720	0.2705
	Impervious	%	0	0.2	0.0005	0.0095	0.0095
	C1	mm	0	5.0	0.736	4.77	2.258
	rec_coef	%	0.01	0.5	0.0517	0.0970	0.0902
	Se_min	mm	50	600	196.9583	571.0728	228.3903
	Ia_coef	%	0.02	0.5	0.0665	0.1927	0.1866

4.1.1. Behavioural parameter distributions

The posterior distributions of the behavioural parameter sets obtained from the Monte Carlo simulations as shown in Figure 2, comparison between the initial uniform parameter ranges and the posterior distributions demonstrates how calibration reduced parameter uncertainty for several model parameters (Beven & Freer, 2001). C1, rec_coef, and AWCper exhibited relatively concentrated behavioural distributions, indicating stronger parameter constraint, whereas Depth, SATper, and Se_min retained broader distributions and therefore greater parameter uncertainty.

The DE-optimized values of C1, rec_coef, AWCper, SATper, and Se_min remained within the ranges supported by the behavioural simulations. The optimized parameter set therefore represents a refinement within the feasible parameter space identified through LHS rather than a departure from the behavioural solutions. Overall, the behavioural distributions demonstrate

heterogeneous parameter uncertainty across the VSA model and provide the basis for the subsequent parameter-identifiability and global sensitivity analyses.

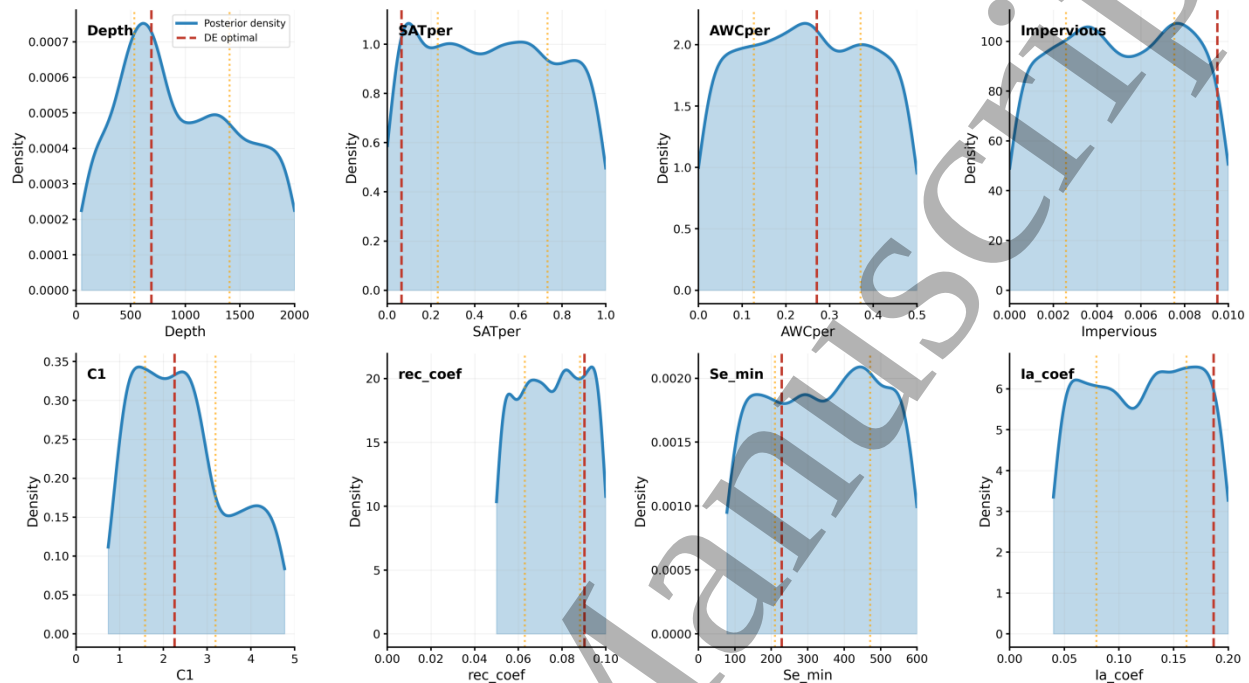


Figure 2: Behavioural parameter distributions obtained from Latin Hypercube Sampling (LHS), with the final optimized parameter values identified by Differential Evolution (DEoptim)

4.1.2. Parameter identifiability and parameter-performance relationships

The parameter-identifiability analysis revealed substantial differences in the degree to which individual VSA parameters were constrained by the observed streamflow record (Figure 3a-c). The Kolmogorov–Smirnov (KS) analysis showed the strongest distributional shifts for Depth and SATper, with C1 also exhibiting a statistically significant shift. In contrast, Impervious, AWCper, rec_coef, Se_min, and Ia_coef showed weaker departures from their prescribed distributions, indicating comparatively weaker constraint by the streamflow data.

The behavioural distributions and DE-optimized values (Figure 3b) showed that the optimized parameter set remained within the behavioural parameter space identified by LHS. The relative position of the optimized values differed among parameters, consistent with their varying degrees of parameter constraint.

The parameter-NSE relationships (Figure 3c) showed the strongest positive association for C1 and the strongest negative association for Depth, while the remaining parameters exhibited relatively weak relationships with NSE. These results indicate that individual parameter-performance relationships provide limited information on overall model behaviour, which reflects the nonlinear and interacting nature of the VSA model. Overall, the results demonstrate heterogeneous parameter identifiability and sensitivity across the VSA parameter space. The KS statistics and parameter-NSE relationships describe how strongly the observed streamflow constrains individual parameters, whereas the subsequent Sobol analysis provides the formal assessment of their global influence on model performance, including interaction effects.

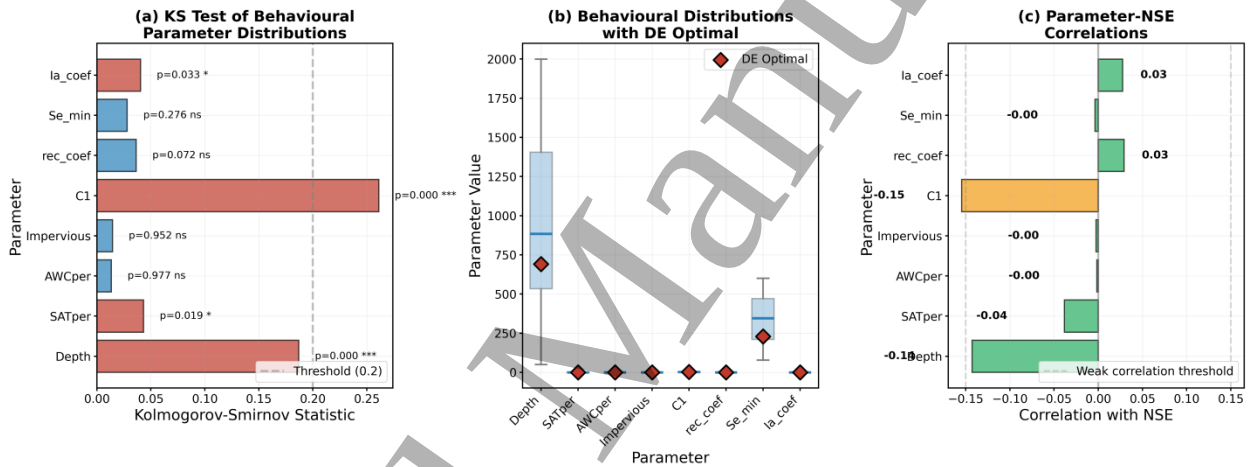


Figure 3: Parameter identifiability based on (a) Kolmogorov–Smirnov (KS) statistics, (b) behavioural distributions and DE optimal, and (c) parameter–NSE correlation.

4.1.3. Global sensitivity analysis (Sobol analysis)

The Sobol analysis revealed a clear hierarchy of parameter influence on VSA model performance (Figure 4; Table 4). The recession coefficient ((rec_coef)) was the dominant parameter, showing substantially greater first and total-order sensitivity than the other parameters. AWCper and Depth exhibited secondary overall influence, whereas Se_min, la_coef, percentImpervious, C1, and SATper showed comparatively low total-order sensitivity.

Table 4: Sobol sensitivity indices of VSA model parameters. (S_1) represents the first-order effect, whereas (S_T) represents the total-order effect including parameter interactions.

Parameter	S_1	95% CI	S_T	95% CI
-----------	-------	--------	-------	--------

rec_coef	0.811	0.789-0.835	0.907	0.819-0.991
AWCper	0.075	-0.005-0.148	0.110	0.096-0.125
Depth	0.033	-0.036-0.109	0.082	0.072-0.094
Se_min	0.010	-0.067-0.082	0.005	0.003-0.006
Ia_coef	<0.001	-0.078-0.076	0.003	0.002-0.004
Impervious	0.001	-0.070-0.077	<0.001	<0.001
C1	0.001	-0.070-0.077	<0.001	<0.001
SATper	0.001	-0.070-0.078	0	0

The results indicate that model performance is particularly sensitive to the recession behaviour represented by (rec_coef), highlighting the importance of representing baseflow recession in the Upper Gilgel Abay watershed. The influence of AWCper and Depth further indicates the importance of soil-water storage processes. Differences between first and total-order indices suggest that interactions contribute to the effects of some parameters. However, parameters with confidence intervals that encompass zero for their first-order indices were not interpreted as having a statistically distinguishable individual effect. The Sobol results complement the parameter-identifiability analysis by distinguishing parameter constraint from parameter influence. A parameter may be relatively well constrained by the observations without being a dominant contributor to output variance, and conversely, a highly influential parameter may remain subject to uncertain. Together, the calibration, identifiability, and global sensitivity analyses provide a more comprehensive assessment of parameter behaviour and model uncertainty.

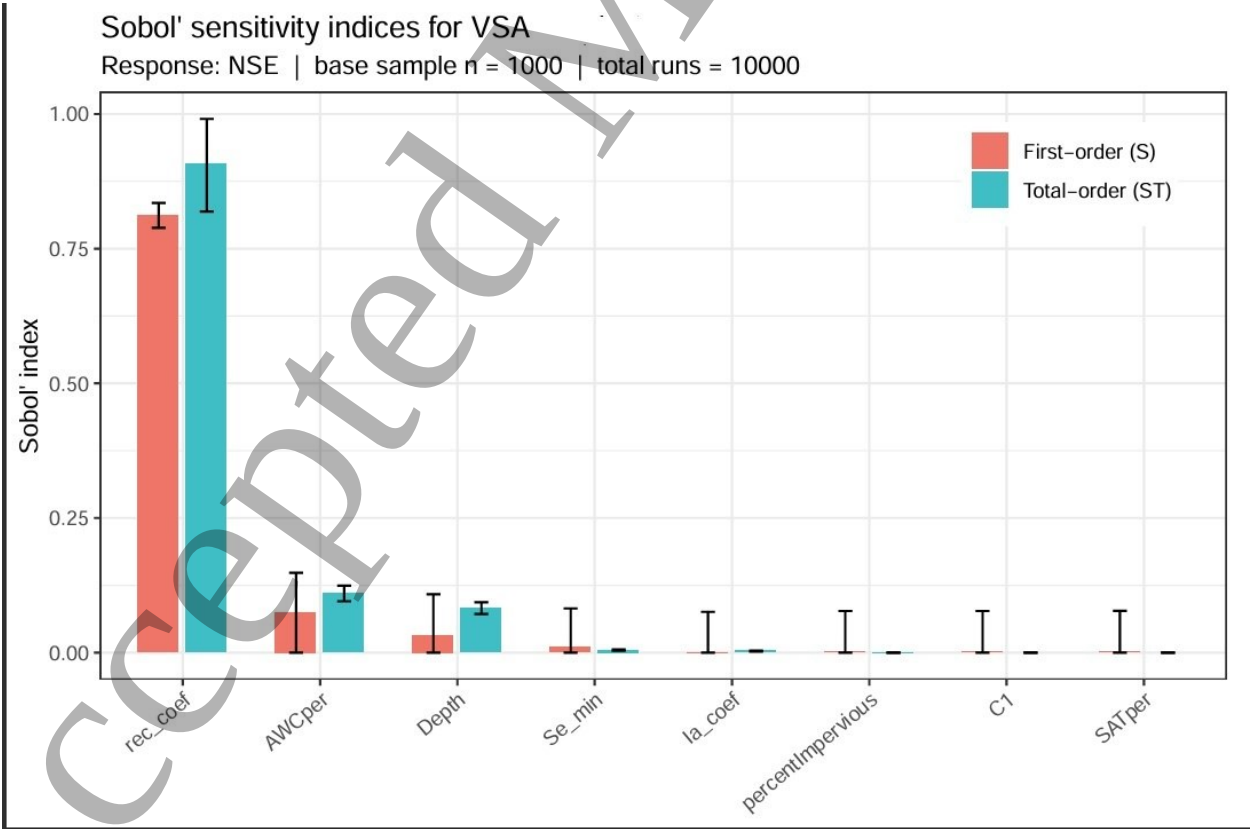


Figure 4: Sobol global sensitivity analysis of VSA model parameters showing first-order (S_1) and total-order (S_T) indices with corresponding confidence intervals. **Note:** (S_1) quantifies the main effect of an individual parameter, whereas (S_T) captures its total effect, including interactions with other parameters.

4.2. Overall model performance

The performance of the VSA model was evaluated during the calibration (1994–2003) and independent validation (2004–2009) periods using NSE, KGE, R^2 , and PBIAS (Table 5). During calibration, the model achieved NSE = 0.811, KGE = 0.86, R^2 = 0.812, and PBIAS = -4.87%, indicating a strong agreement between observed and simulated daily streamflow. During the independent validation period, the model achieved NSE = 0.69, KGE = 0.75, R^2 = 0.80, and PBIAS = 5.13%.

Table 5: Performance statistics of the VSA model during calibration (1994–2003) and validation (2004–2009)

Performance Metric	Calibration (1994–2003)	Validation (2004–2009)
NSE	0.811	0.69
KGE	0.86	0.75
R^2	0.812	0.80
PBIAS%	-4.87	5.13

The NSE and KGE values decreased from the calibration to the validation period, indicating some reduction in predictive performance under independent hydro-climatic conditions. Nevertheless, the validation results remained strong, with an NSE close to 0.70 and KGE above 0.75. The relatively high R^2 values during both calibration and validation (0.812 and 0.80, respectively) indicate a strong linear correspondence between observed and simulated streamflow. The relatively small PBIAS values (-4.87% during calibration and 5.13% during validation) indicate limited overall volumetric bias, with the negative calibration value indicating slight underestimation and the positive validation value indicating slight overestimation.

Overall, the results demonstrate that the two-stage LHS-DEoptim calibration framework produced a VSA parameter set capable of reproducing daily streamflow with good consistency between the calibration and independent validation periods.

4.3. Streamflow simulation

4.3.1. Hydrograph comparison

The performance of the Variable Source Area (VSA) model in simulating streamflow is evaluated against observed discharges for both the calibration (1994-2003) and validation (2004-2009) periods, as shown in Figure 5. Overall, the model successfully reproduces the temporal variability and seasonal runoff patterns characteristic of the Upper Gilbel Abay watershed.

During the calibration period (1994–2003), simulated streamflow closely follows the observed hydrograph, reproducing the timing of most runoff events and seasonal recession patterns. Peak flows are generally captured satisfactorily, although a few extreme events are slightly underestimated. In the validation period (2004-2009), the model continued to simulate the major seasonal dynamics and the timing and magnitude of most runoff events. The later validation years should nevertheless be interpreted with some caution, as previous studies have reported uncertainties and potential quality issues in hydrometric records in the Upper Blue Nile Basin, including rating-curve and gauge-related problems (Negatu et al., 2022; Teferi et al., 2025; Wale et al., 2009). These data-related uncertainties may have contributed to some differences between observed and simulated discharge during the validation period.

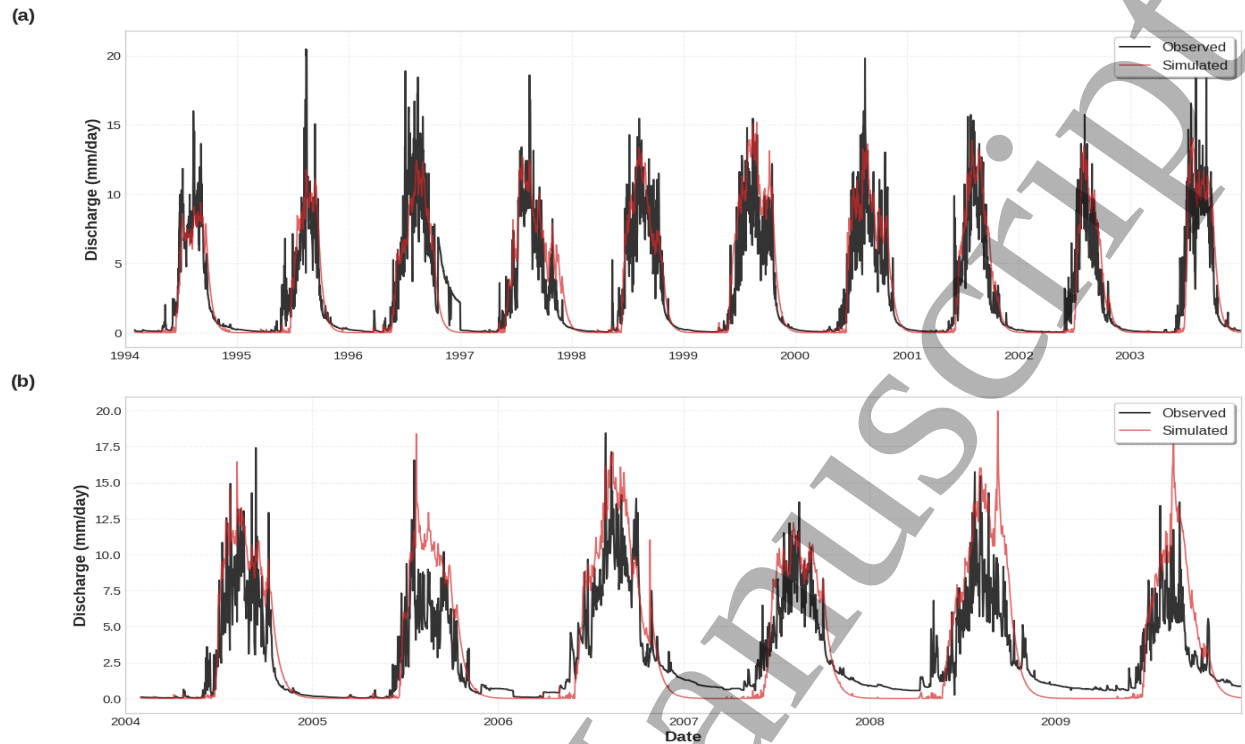


Figure 5: Comparison of observed and VSA-simulated daily streamflow hydrographs for the Upper Gilgel Abay watershed: (a) calibration period (1994-2003) and (b) validation period (2004-2009).

4.3.2. Scatter plot analysis

The relationship between observed and simulated daily streamflow was further examined using scatter plots and linear regression (Figure 6). The simulated and observed flows showed strong linear associations during both calibration and validation, with coefficients of determination (R^2) of approximately 0.81 and 0.80 respectively. The scatter plots show some deviation from the 1:1 relationship, particularly at higher flows, with differences also evident during validation. Overall, the results indicate a strong correspondence between observed and simulated daily streamflow, while the NSE, KGE, and PBIAS values provide the primary assessment of model performance.

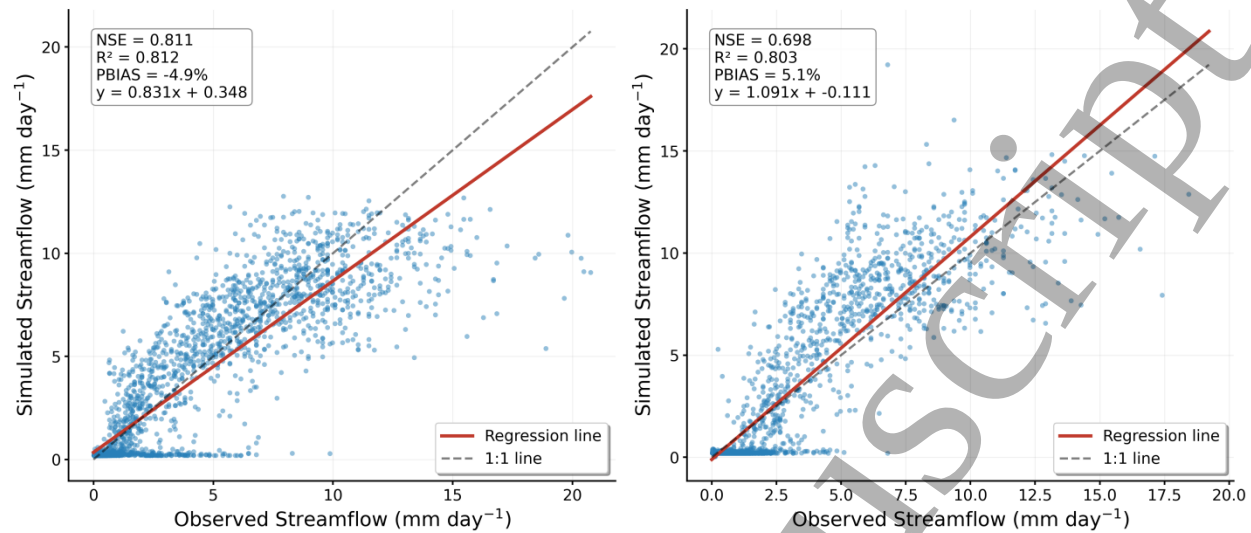


Figure 6: Scatter plots comparing observed and VSA-simulated daily streamflow during the (a) calibration period (1994-2003) and (b) validation period (2004-2009).

4.3.3. Flow duration curve

The flow duration curve (Figure 7) provides a comprehensive visual assessment of the VSA model's performance across the entire spectrum of daily streamflow conditions in the Upper Gilgel Abay watershed. The agreement is particularly evident at exceedance probabilities below approximately 50%, indicating that the model reproduces the overall magnitude and variability of moderate- to high-flow conditions.

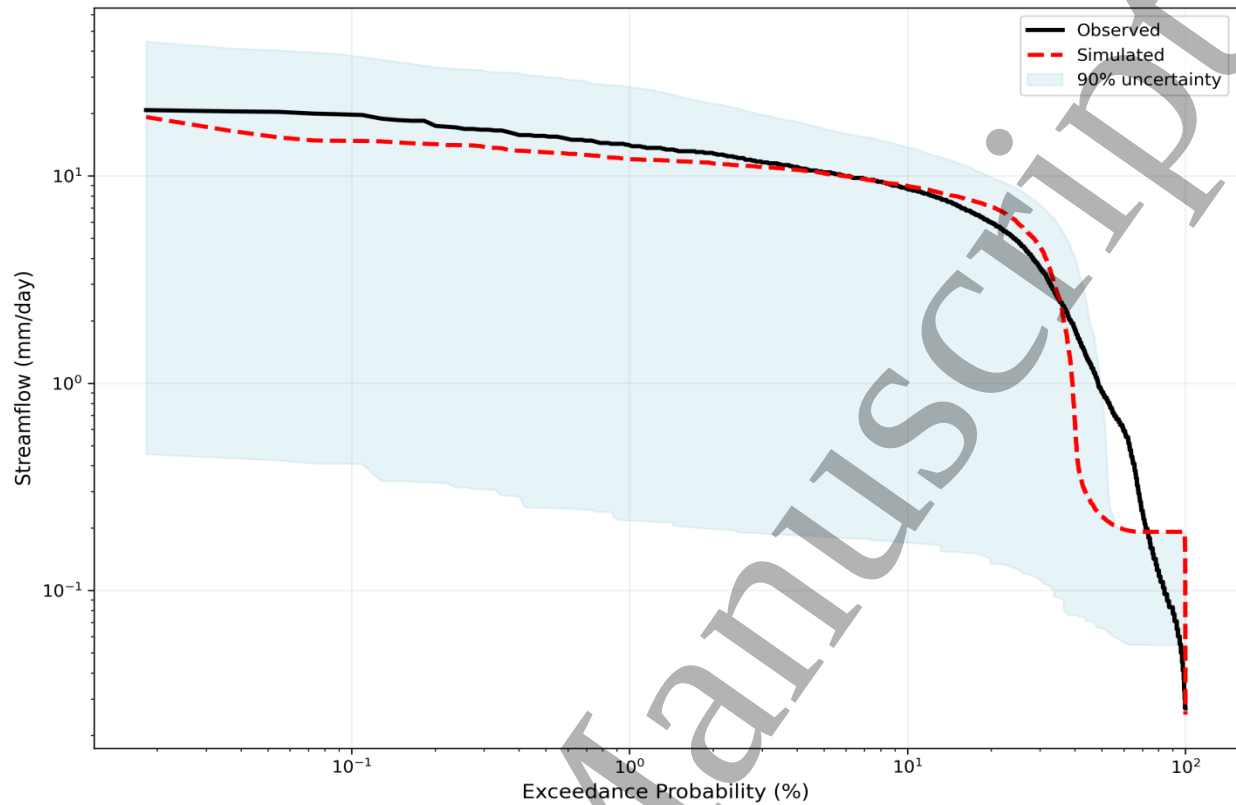


Figure 7: Flow duration curves of observed and simulated daily streamflow for the Upper Gilgel Abay watershed during 1994-2009

At higher exceedance probabilities, representing lower-flow conditions, the simulated discharge increasingly falls below the observed values, indicating some underestimation of sustained low flows. This pattern is consistent with the hydrograph and simulated flow components, which show differences during prolonged low-flow and recession periods. The 90% prediction uncertainty envelope encompasses a substantial portion of the observed flow-duration curve, particularly across intermediate and high-flow conditions, although differences remain in parts of the low-flow range. Overall, the flow-duration analysis indicates that the calibrated VSA model simulates the main characteristics of the observed flow regime, with relatively discrepancies under low-flow conditions.

4.4. Hydrological process representation

4.4.1. Simulated streamflow components

Simulated streamflow was partitioned into baseflow, interflow, and overland flow to examine the temporal variability of the modelled flow components (Figure 8). The components exhibited distinct seasonal patterns. Overland flow and interflow increased during the wet season and coincided with periods of higher simulated streamflow, whereas baseflow remained comparatively smaller and more persistent throughout the simulation period.

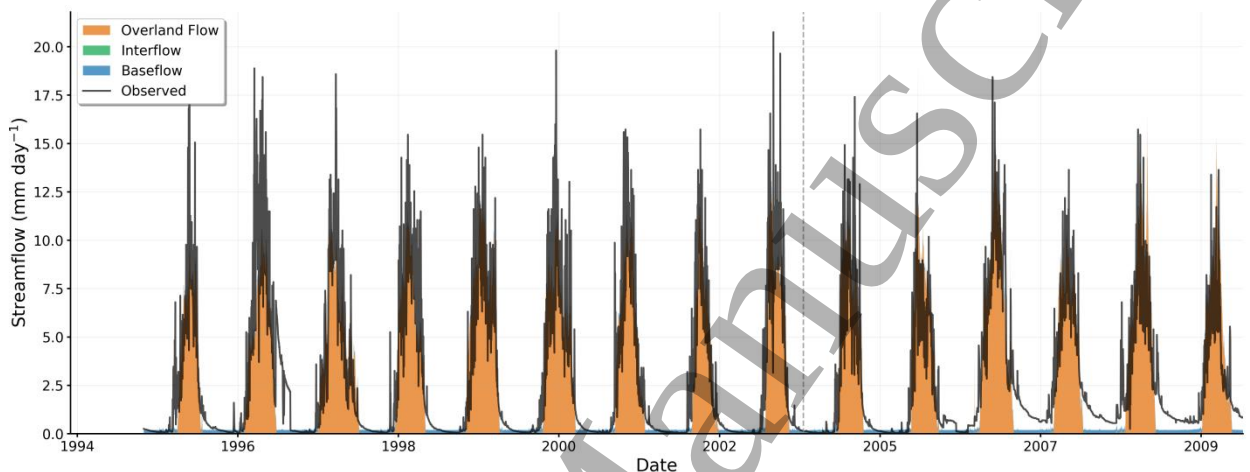


Figure 8: Simulated daily streamflow components in the Upper Gilgel Abay watershed during the calibration (1994–2003) and validation (2004–2009) periods, (*Contributions from baseflow, shallow interflow, and overland flow*).

Overland flow was most pronounced around seasonal flow peaks, while interflow provided an additional subsurface contribution during wetter periods. Baseflow became proportionally more prominent during recession and dry-season periods as the rapid runoff components declined. Overall, the simulated flow partitioning indicates greater contributions from overland flow and interflow during wet periods, with a persistent baseflow contribution during lower-flow conditions. Because these components are internal model states and were not independently measured, they are interpreted as model-derived evidence of simulated runoff partitioning rather than direct observational confirmation.

4.4.2. Saturated source area dynamics

The VSA model simulated the temporal variation in saturated source-area fraction throughout the study period (Figure 9). The saturated area generally expanded during the wet season (June–

September) and contracted during the dry season. The 30-day moving average shows a recurring seasonal pattern across both the calibration and validation periods.

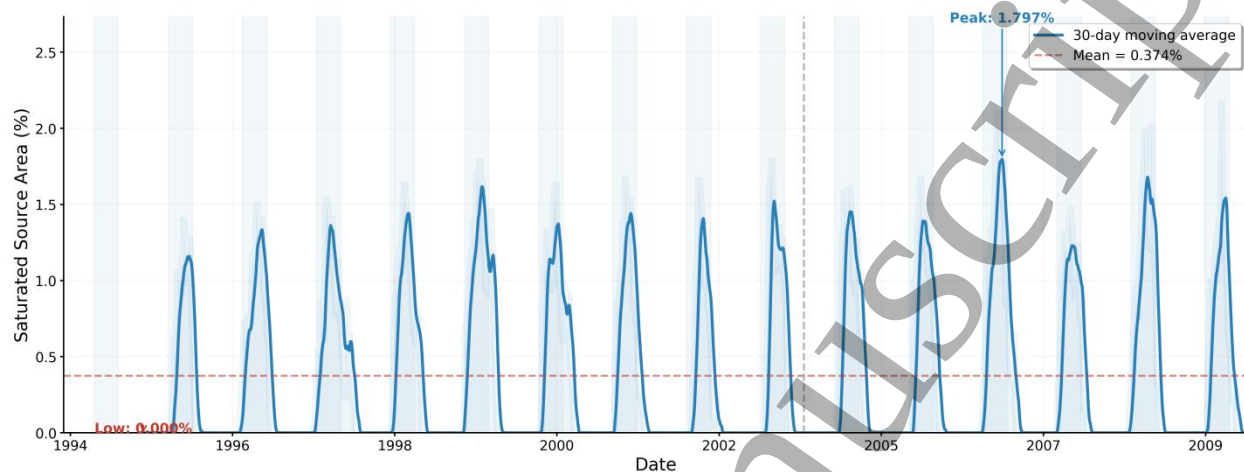


Figure 9: Temporal variation of simulated saturated source area fraction in the Upper Gilgel Abay watershed during the calibration (1994-2003) and validation (2004-2009) periods.

The simulated saturated-area fraction was highest mainly during August-September and lowest during the dry season. The seasonal maximum and minimum were approximately 1.30% and 0.40%, respectively, corresponding to an approximately 3.25-fold difference in the simulated contributing area. Overall, the results show pronounced seasonal variability in the simulated saturated source area, with larger contributing areas during wetter periods and smaller areas during drier conditions.

4.4.3. Soil water storage dynamics

The simulated soil water and effective storage exhibit consistent seasonal dynamics across the calibration (1994-2003) and validation (2004-2009) periods (Figure 10). Soil water storage generally declines during the dry season and increases with the onset of the main rainy season, reaching higher levels during the wettest months before declining again toward the end of the rainy season.

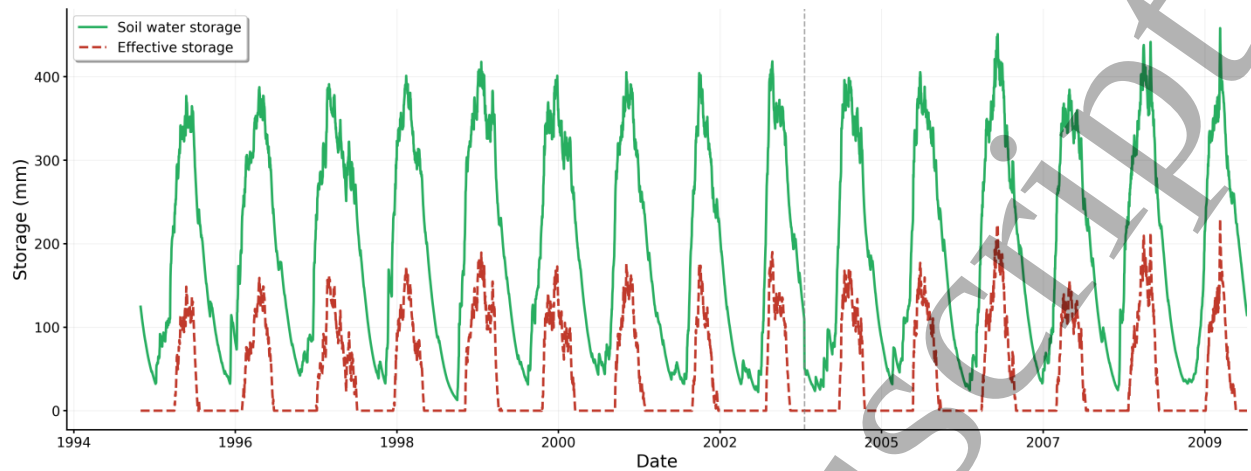


Figure 10: Temporal variation of simulated soil water storage and effective storage in the Upper Gilgel Abay watershed during the calibration (1994-2003) and validation (2004-2009) periods.

Effective storage shows a similar seasonal pattern but responds more strongly to changes in catchment wetness. It increases during periods of higher soil water storage and declines during the subsequent drying period. These patterns are broadly consistent between the calibration and validation periods, indicating that the model reproduces the seasonal wetting and drying dynamics of the catchment. Overall, the simulated storage variables show clear seasonal variability, with higher storage during the rainy season and progressive depletion during the dry season.

4.5. Monthly water balance

The mean monthly rainfall, evapotranspiration, and simulated streamflow for 1994-2009 are presented in Figure 11. Rainfall exhibits a distinct unimodal seasonal pattern, with low values during the dry months and a rapid increase from March onward. Rainfall reaches maximum mean daily values in July (13.4 mm/day) and August (13.2 mm/day), with June-September representing the main rainy season.

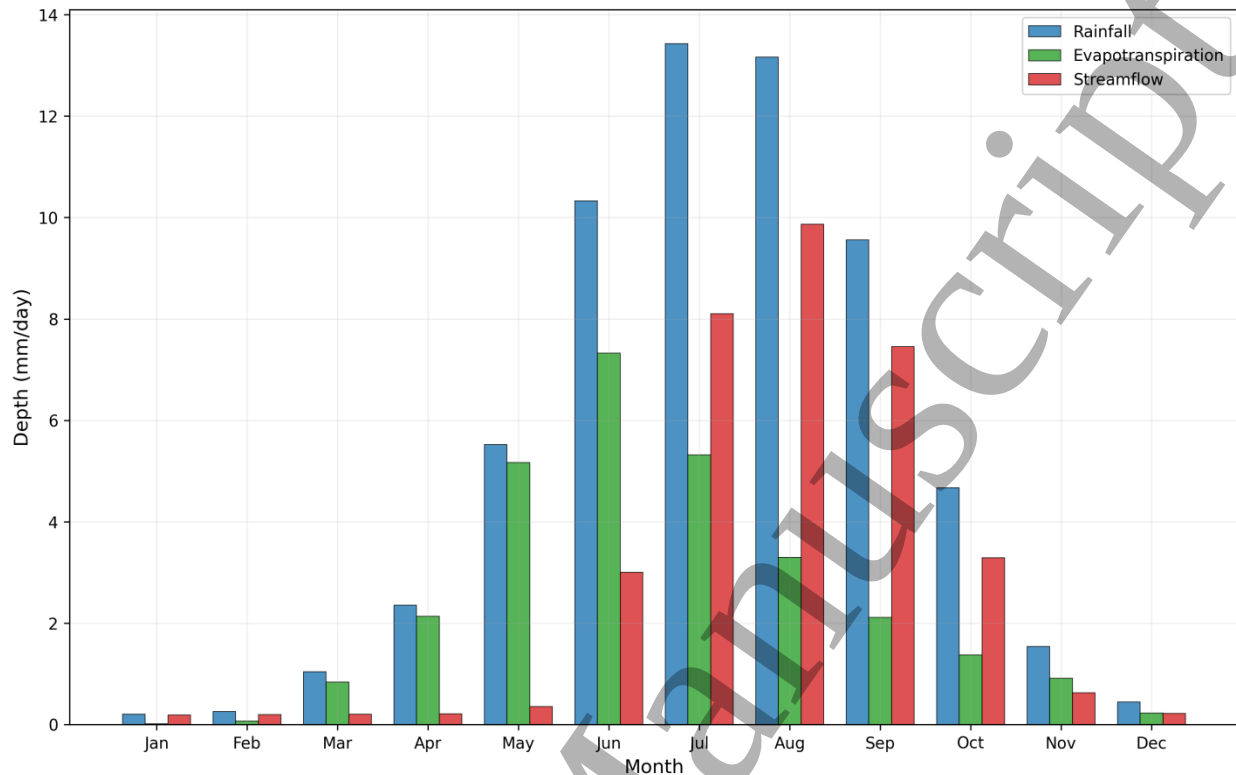


Figure 11: Mean monthly rainfall, evapotranspiration, and simulated streamflow during 1994-2009.

Simulated streamflow follows the seasonal rainfall pattern, with low values during the dry season (≤ 0.2 mm/day) and a rapid increase from June. Streamflow reaches a maximum of 9.9 mm/day in August, shortly after the rainfall peak. The seasonal difference between rainfall and streamflow coincides with increased evapotranspiration and changes in soil-water storage (Figure 10). Simulated evapotranspiration also varies seasonally, increasing from 2.1 mm/day in April to a maximum of 7.4 mm/day in June, before declining toward the end of the rainy season. Overall, the simulated water-balance components show a clear seasonal response, with rainfall, evapotranspiration, and streamflow varying consistently with the annual hydro-climatic cycle.

4.6. Model uncertainty analysis using GLUE

Predictive uncertainty of the calibrated VSA model was assessed using the behavioural parameter ensemble within the GLUE framework (Figure 12). The figure shows observed daily streamflow, the GLUE-median simulation, and the 90% prediction uncertainty band for 1994-2009. The GLUE-median simulation generally reproduces the seasonal pattern of observed

streamflow, including wet-season peaks and dry-season low flows. The 90% prediction uncertainty band contains 69.3% of the observed streamflow values, indicating substantial coverage of the observed hydrograph by the behavioural parameter ensemble. The uncertainty band is generally wider during higher-flow periods and narrower during low-flow conditions.

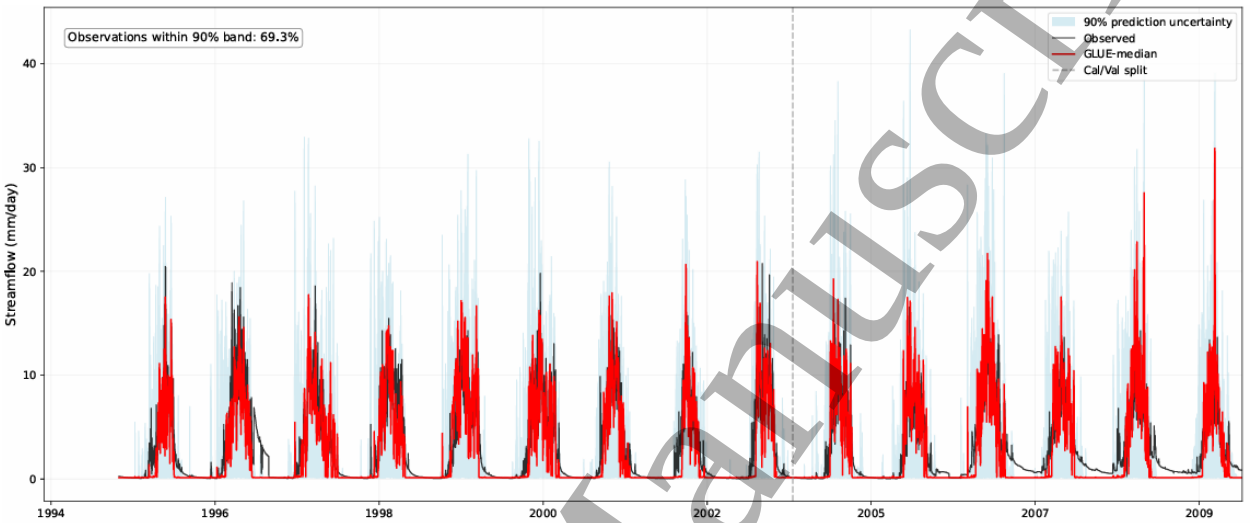


Figure 12: Observed versus GLUE-median simulated daily streamflow (mm/day) with the 90% prediction uncertainty band for the Upper Gilgel Abay watershed (1994-2009).

Some differences remain during individual high-flow events and recession periods, particularly during validation. Overall, the GLUE results demonstrate that the behavioural parameter ensemble provides substantial predictive uncertainty coverage across the study period.

5. Discussion

5.1. Overall model performance

The VSA hydrological model successfully simulated daily streamflow dynamics in the Upper Gilgel Abay watershed despite the limited availability of hydrological and meteorological data. The calibrated VSA model achieved very good performance during calibration (NSE = 0.811, KGE = 0.865, PBIAS = -4.87%) and good performance during independent validation (NSE = 0.697, KGE = 0.754, PBIAS = 5.13%). These results demonstrate that the calibrated parameter set retained reasonable predictive capability under independent climatic conditions. The relatively small change in PBIAS between calibration and validation indicates that the model did not exhibit a strong systematic bias when transferred to an independent period. The reduction in

NSE and KGE during validation nevertheless indicates that some aspects of the hydrological response were not fully transferable between the two periods. According to the performance guidelines of Moriasi et al. (2007), these statistics indicate that the VSA model is suitable for daily streamflow simulation in data-scarce watersheds.

The hydrograph comparison (Figure 5) demonstrates good agreement between observed and simulated streamflow during both calibration and validation periods, particularly regarding the timing of seasonal runoff events. The scatter plot further confirms that simulated streamflow closely follows observed discharge over a wide range of flow conditions. Although some dispersion remains during extreme flood events, the overall correspondence suggests that the VSA framework adequately represents the dominant hydrological behaviour of the watershed.

The calibration strategy also contributed to the identification of a physically plausible parameter set. Latin Hypercube Sampling provided an efficient exploration of the multidimensional parameter space and identified behavioural parameter combinations, while Differential Evolution subsequently refined the solution within the feasible parameter domain. Importantly, the optimized solution remained within the behavioural parameter space identified during the initial sampling stage. This provides evidence that the final parameter set was not an isolated mathematical optimum outside the range of hydrologically acceptable solutions.

The obtained performance is comparable with the performance reported for conceptual rainfall-runoff models applied in data-limited catchments of the Upper Blue Nile Basin (Bizuneh et al., 2021; Setegn et al., 2010). However, the principal value of the present application is not solely the predictive accuracy of the model. The VSA framework additionally provides a process-based interpretation of runoff generation through the expansion and contraction of saturated source areas. This distinction is important in the humid Ethiopian Highlands, where saturation-excess runoff can play a major role in streamflow generation (Easton et al., 2008; Steenhuis et al., 2009; Zimale et al., 2017).

5.2. Hydrological controls revealed by parameter identifiability and sensitivity

The behavioural and KS analyses showed substantial differences in parameter constraint. Depth, SAT_{per}, and C1 exhibited the strongest distributional shifts between the prescribed and behavioural parameter spaces, whereas Impervious, AWC_{per}, and Ia_{coef} remained

comparatively broad. These results indicate that streamflow observations provide stronger information for constraining some storage and saturation-related parameters than others. The broad behavioural ranges are also consistent with parameter sensitivity, whereby multiple parameter combinations can produce similarly acceptable simulations (Beven, 2006).

The parameter-NSE relationships provided a complementary assessment, with C1 showing the strongest positive association and Depth the strongest negative association, while most other parameters exhibited weak relationships. These associations should not be interpreted as formal sensitivity measures. Identifiability reflects how strongly observations constrain parameter values, whereas sensitivity describes how changes in parameter values influence model output.

The Sobol analysis provided the formal global sensitivity assessment. `rec_coef` showed the greatest influence on model-output variability, followed by `AWCper` and `Depth`. The dominant influence of `rec_coef` is hydrologically consistent with its role in controlling groundwater recession and the persistence of streamflow during inter-event and dry-season periods. The sensitivity of `AWCper` and `Depth` further highlights the importance of soil-water storage in regulating the transition toward saturation and runoff generation.

Comparison of the first-order (S_1) and total-order (ST) Sobol indices provides additional information on parameter interactions. Where ST substantially exceeded S_1 , the parameter's influence was partly associated with interactions with other parameters rather than its individual effect alone. Conversely, a smaller difference between S_1 and ST indicates a greater contribution from the parameter's direct effect. Thus, the Sobol results demonstrate that model-output variability reflects both individual parameter effects and parameter interactions.

Taken together, the analyses distinguish parameter sensitivity from parameter influence: KS and behavioural distributions identify how strongly streamflow observations constrain parameters, parameter-NSE relationships describe their association with model performance, and Sobol indices quantify their global influence and interactions. Overall, groundwater recession and soil-water storage emerged as the principal controls on simulated streamflow variability, while several other parameters remained less sensitive to discharge observations alone. The results therefore highlight the value of additional groundwater, soil-moisture, and catchment-storage observations for reducing uncertainty in future applications.

5.3. Hydrological process representation

The simulated hydrographs, flow components, soil-water storage, and saturated source-area dynamics collectively indicate that catchment wetness plays an important role in the simulated runoff response. During the wet season, increasing soil-water storage was accompanied by expansion of saturated source areas and greater contributions from overland flow and interflow, followed by contraction of saturated areas and increasing relative importance of baseflow during recession periods. This seasonal behaviour is consistent with the conceptual basis of Variable Source Area (VSA) theory, in which runoff-producing areas expand as catchment wetness increases and contract as storage is depleted (Panjabi et al., 2025; Steenhuis et al., 1995).

The results therefore suggest that rainfall magnitude alone does not determine the simulated runoff response. Antecedent wetness regulates how much of the catchment approaches saturation and consequently influences the extent of runoff-producing areas. Thus, rainfall events of similar magnitude may generate different streamflow responses under different antecedent moisture conditions. This process-based representation is a key feature of the VSA framework and provides a useful basis for interpreting runoff generation in humid highland catchments, where saturation-excess processes have been reported as important (Bayabil et al., 2010; Tilahun et al., 2016).

The model-derived flow components further indicate a shift in runoff partitioning across the seasonal cycle. Overland flow and interflow increased during wetter periods, whereas baseflow provided a more persistent contribution during recession and dry-season conditions. Because these components and saturated source areas are internal model states rather than independently observed quantities, they should be interpreted as process-based model evidence rather than direct observational confirmation.

The persistent underestimation of some low-flow periods is particularly relevant in light of the strong Sobol sensitivity of `rec_coef`. In the VSA formulation, this parameter controls groundwater recession, whereas actual groundwater behaviour in the Ethiopian Highlands may involve heterogeneous weathered materials, fractured bedrock, localized groundwater barriers, and multiple subsurface flow pathways (Collick et al., 2015). Such heterogeneity can produce delayed and multi-timescale groundwater release that is difficult to represent with a single

recession coefficient. Evidence from Ethiopian catchments also indicates that geological structure, bedrock permeability, and weathered-regolith characteristics can substantially influence groundwater recharge and baseflow behavior (Sishu et al., 2024; Finsa and Bruthans., 2025).

These findings suggest that the remaining low-flow discrepancies may partly reflect hydrogeological processes that are simplified in the current model structure. More explicit representation of multiple groundwater reservoirs or delayed subsurface pathways could potentially improve recession simulation, but such extensions would require additional groundwater observations to constrain the associated parameters. Overall, the results support the usefulness of the VSA framework as a parsimonious representation of wetness-dependent runoff generation, while highlighting the limitations associated with simplified groundwater processes.

5.4. Validation behaviour, predictive uncertainty and flow-regime representation

The independent validation showed a moderate reduction in NSE and KGE relative to calibration, while PBIAS remained relatively low (5.13%). This indicates that the calibrated model retained reasonable predictive capability, although some aspects of the rainfall storage-runoff relationship were less consistently simulated during 2004-2009. Differences in rainfall characteristics, seasonality, event magnitude, and antecedent moisture conditions between the two periods may have contributed to this reduction in performance (Fowler et al., 2018; Shen, 2023). Uncertainty in both precipitation inputs and observed discharge may also contribute to differences between simulated and observed streamflow, particularly in data-scarce catchments (Guo et al., 2020). Uncertainty in the discharge observations is particularly relevant when interpreting individual validation events. Previous studies have reported uncertainties and potential quality issues in hydrometric records in the Upper Blue Nile Basin, including rating-curve and gauge-related problems (Negatu et al., 2022; Teferi et al., 2025; Wale et al., 2009). Such uncertainties may contribute to some differences between observed and simulated discharge, particularly during high-flow events, and therefore the reduction in validation performance should not be attributed solely to model structural limitations.

The GLUE analysis provides a complementary assessment of predictive uncertainty. The 90% prediction interval enclosed 69.3% of the observed streamflow values, indicating that the

behavioural parameter ensemble captured a substantial portion of the observed variability. However, the coverage below the nominal 90% level indicates that parameter uncertainty alone does not fully account for the discrepancies between simulations and observations. This suggests that input-data and structural uncertainties also contribute to predictive uncertainty.

The uncertainty envelope was generally wider during high-flow conditions, reflecting greater variability among behavioural simulations during periods of strong runoff response. During recession and low-flow periods, uncertainty was more closely related to the simplified representation of groundwater recession, consistent with the strong Sobol sensitivity of *rec_coef*. Thus, the combined GLUE, parameter-identifiability, and Sobol results indicate that predictive uncertainty arises from both parameter variation and limitations in representing key hydrological processes. Overall, the validation results demonstrate that the VSA model maintains reasonable predictive capability under independent conditions, while also highlighting uncertainty in both model structure and observations. The GLUE results complement deterministic performance metrics by providing a range of plausible streamflow responses rather than relying solely on the single optimized parameter set.

5.5. Watershed management implications

The results have important implications for hydrological management in the Upper Gilgel Abay watershed and the wider Upper Blue Nile Basin. The parameter-identifiability and Sobol analyses indicate that soil-water storage and groundwater recession are important controls on simulated streamflow variability. This highlights the importance of management approaches that consider soil-water retention, infiltration, groundwater storage, and the timing of runoff generation rather than focusing only on rainfall-driven peak flows.

The VSA framework further indicates that runoff-producing areas vary with catchment wetness rather than remaining spatially and temporally fixed. This behaviour is consistent with the Variable Source Area theory originally proposed by Dunne and Black (1970). During wetter periods, expansion of saturated areas can increase runoff generation and connectivity, whereas contraction during drier periods reduces the area contributing rapid runoff. Identifying locations and periods with greater saturation potential could therefore support more targeted soil and water conservation, erosion-control, agricultural water management, and flood-risk assessment

(Tilahun et al., 2016). Such an approach is more consistent with the dynamic nature of runoff generation than treating the watershed as hydrologically uniform.

The strong sensitivity of *rec_coef* also highlights the importance of groundwater recession and delayed subsurface contributions to maintaining streamflow during dry periods. Measures that enhance soil-water retention and infiltration may have potential benefits for both reducing rapid runoff and supporting subsurface storage; however, these implications should be interpreted cautiously because the current model does not explicitly resolve spatial groundwater processes. Additional hydrogeological information would therefore be required before translating the model results into site-specific groundwater-management interventions.

The results also suggest that changes in antecedent wetness and rainfall variability may alter the balance between rapid runoff and delayed flow. Wetter conditions may increase the extent of saturated source areas and consequently enhance rapid runoff, whereas prolonged dry conditions may increase uncertainty in simulated low flows because of limitations in the groundwater representation. The VSA framework can therefore provide a useful process-based basis for exploring watershed-management and climate-related scenarios, particularly in data-scarce settings. Overall, the combined behavioural, identifiability, Sobol sensitivity, and GLUE analyses provide a basis for considering hydrological predictions as a range of plausible responses rather than a single deterministic outcome. Future monitoring should prioritize groundwater dynamics, soil-water storage, and reliable high-flow discharge measurements, as these observations would help constrain the processes identified as important sources of model uncertainty.

5.6. Limitations and future research

Although the VSA model successfully represented the dominant saturation-excess runoff processes in the Upper Gilgel Abay watershed, these limitations should be acknowledged. Some storage-related parameters were weakly constrained by streamflow observations despite the combined behavioural, identifiability, Sobol sensitivity, and GLUE analyses, indicating persistent parameter uncertainty. The simplified groundwater representation and semi-distributed structure may also limit the representation of complex subsurface processes and spatial variability in soil, land use, geology, and groundwater dynamics, particularly during low-flow

periods. In addition, potential evapotranspiration was estimated using a temperature-based method because complete meteorological data required for the FAO Penman-Monteith approach were unavailable, introducing additional input uncertainty (Allen et al., 1998). Although seasonal hydrological behaviour was evaluated using catchment storage, saturated source-area dynamics, flow components, hydrographs, and flow-duration characteristics, separate wet- and dry-season performance statistics were not calculated and therefore remain an area for further assessment. Future research should incorporate improved hydrogeological and soil-moisture observations, more spatially explicit groundwater processes, and improved PET estimation. Evaluation under future climate and land-use scenarios would further enhance the applicability of the VSA framework for adaptive water-resources management.

6. Conclusion

This study evaluated the applicability of the Variable Source Area (VSA) hydrological model for simulating daily streamflow in the Upper Gilgel Abay watershed using a two-stage calibration framework combining Latin Hypercube Sampling (LHS) and Differential Evolution optimization (DEoptim). The LHS stage efficiently explored the feasible parameter space and identified behavioural parameter sets, whereas DE optimization refined these behavioural ranges to obtain the optimum parameter set. The calibrated model achieved very good performance during calibration ($NSE = 0.811$, $KGE = 0.865$, $PBIAS = -4.87\%$) and good performance during independent validation ($NSE = 0.697$, $KGE = 0.754$, $PBIAS = 5.13\%$), demonstrating that the VSA model provides reliable streamflow simulations under data-scarce conditions. The combined uncertainty, identifiability, and sensitivity analyses revealed substantial differences among model parameters. The Sobol analysis identified rec_coef as the dominant parameter controlling model performance, with AWC_{per} and $Depth$ showing secondary influence. Behavioural parameter distributions further indicated that storage-related parameters, particularly $Depth$ and SAT_{per} , remained comparatively uncertain, demonstrating that streamflow observations alone provide limited constraints on some internal storage processes. These results highlight the importance of groundwater recession and antecedent soil-water conditions in controlling runoff generation within the watershed. Hydrograph, flow-duration-curve, saturated-source-area, and internal-storage analyses further demonstrated that the VSA model reproduced the major seasonal dynamics of the watershed and provided a physically consistent

representation of saturation-excess runoff generation. Nevertheless, occasional peak-flow overestimation and low-flow underestimation indicate limitations associated with simplified groundwater representation, hydrogeological complexity, and uncertainties in hydro-meteorological observations. Overall, this study provides the first application of the VSA framework to the Upper Gilgel Abay watershed and demonstrates its potential for both streamflow prediction and interpretation of runoff-generation processes in data-scarce Ethiopian highland catchments. The integration of LHS, DEoptim, parameter uncertainty and identifiability analysis, and Sobol sensitivity analysis provides a robust framework for parameter estimation and model diagnosis. Future research should focus on improving groundwater representation, incorporating spatially distributed soil, geological, and remotely sensed information, and evaluating model robustness under changing hydro-climatic conditions.

Author's contribution

AGW: Writing - review and editing, writing - first draft, Idea generation, Technique, Formal investigation, and Presentation of findings. FAZ: Writing -review and editing, Methodology, Guidance, and Idea generation.

Declarations

Ethics and Consent to Participate: not applicable

Funding: No funding

Data availability: Data will be provided upon request.

Competing interests: No competing Interests

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